

D855

Natural Hazards

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A coastal nation with a fast-growing tourism industry sits where a dense oceanic plate sinks beneath a continental plate. Its capital has a new subway system. Which set of hazards should its planners expect?

- A. Earthquakes but no volcanoes.
- B. Gentle basaltic eruptions and small, shallow earthquakes.
- C. Large megathrust earthquakes, tsunamis, and explosive volcanoes.
- D. Tsunamis only, since the subduction is under water.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A region has frequent shallow earthquakes along a long fault where two plates slide past each other horizontally, but it has no volcanoes. What kind of plate boundary is this?

- A. A divergent boundary.
- B. A transform boundary.
- C. A transform boundary, which always produces large tsunamis.
- D. A subduction (convergent) boundary.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A seismic network with 120 stations records a magnitude 7.0 earthquake. A mountain village near the epicenter reports heavy damage (intensity IX), while a city 300 km away that hosts a university reports light shaking (intensity IV). How should these numbers be understood?

- A. The quake had two magnitudes, IX and IV.
- B. One of the reports must be wrong, since intensity depends only on magnitude.
- C. The quake has one magnitude; intensity differs by place, mainly with distance and local ground.
- D. Intensity is always the same everywhere for a single quake.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

About how much more energy does a magnitude 8.0 earthquake release than a magnitude 6.0 earthquake?

- A.** 64 times
- B.** 2 times
- C.** 100 times
- D.** About 1,000 times

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A volcano observatory staffed by 15 scientists monitors two volcanoes. Volcano X erupts low-silica basalt; Volcano Y erupts high-silica rhyolite rich in dissolved gas. Volcano X is closer to the observatory's access road. Which is more likely to erupt explosively?

- A.** Both equally, since explosivity depends only on volcano size.
- B.** Volcano Y, because its thick, sticky magma traps gas until it escapes violently.
- C.** Volcano X, because basalt builds the largest volcanoes.
- D.** Volcano X, because hotter basalt is more violent.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

Residents of a fishing town with a colorful harbor festival feel a long, strong earthquake. Minutes later the sea pulls far back from the beach, exposing the seafloor. A tour boat is scheduled to leave at noon. What should residents do?

- A.** Wait for an official siren, since natural signs are never reliable.
- B.** Go to high ground but return after the first wave, since a tsunami is one wave.
- C.** Walk out to collect stranded fish before the water slowly returns.
- D.** Move to high ground or inland immediately and stay there until officials give the all-clear.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A hurricane with 190 km/h winds, named by the regional warning center, crosses 29 °C ocean water and then moves over a cold current of 22 °C. Its forecast track passes near a major shipping port. What will most likely happen to its strength?

- A. It strengthens, because the colder air adds energy.
- B. It weakens, because cool water cannot supply enough heat and moisture.
- C. It weakens because the Coriolis effect stops over cold water.
- D. Its strength stays fixed once it has formed.

Target word _____	The one fact that decides _____	Answer ____	[] sure [] unsure [] guess
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8.

A town of 9,000 is built on the floodplain of a river whose 100-year flood reached the courthouse steps in 1998. The town's bridge was repainted last year. What is the probability that a flood at least that large occurs in any given year?

- A. 1%
- B. 99%
- C. It is certain to occur exactly once every 100 years.
- D. About 0%, since the next one is due around 2098.

Target word _____	The one fact that decides _____	Answer ____	[] sure [] unsure [] guess
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9. [TRANSFER — reversed stem / different variable]

A 50-year flood has a 2% chance of happening in any year. What is the chance of at least one such flood during a 30-year mortgage?

- A. 2%
- B. 60%
- C. About 55%
- D. About 45%

Target word _____	The one fact that decides _____	Answer ____	[] sure [] unsure [] guess
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10.

A county that promotes its farmers' markets has paved 40% of a once-forested watershed for new subdivisions over 20 years. The same size rainstorm now produces a higher and faster river crest. The county seat's library was recently expanded. Why?

- A. Land use cannot change floods; only rainfall can.
- B. The subdivisions raised evaporation, which added water to the river.
- C. Pavement and storm drains reduce infiltration and speed runoff to the river.
- D. Paving increases infiltration, so groundwater floods the river.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

11. [TRANSFER — reversed stem / different variable]

A region that once had about one felt earthquake a year begins having dozens a year after large volumes of wastewater are injected deep underground. Which explanation fits best?

- A. The injected fluid raises pressure on existing faults, reducing friction so they slip.
- B. The injection is creating a new plate boundary.
- C. It is coincidence, since people can never cause earthquakes.
- D. Pumping water out lets the rocks collapse.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

12.

Two towns sit the same distance from the same active fault and face the same expected shaking. Town P has 50,000 people in unreinforced brick buildings; Town Q has 50,000 people in buildings retrofitted to modern codes. Town P has a famous clock tower. Which statement is correct?

- A. The hazard is the same, but Town P faces higher risk because its buildings are more vulnerable.
- B. Town P faces a greater earthquake hazard.
- C. Both towns face the same risk because they face the same hazard.
- D. Town Q faces higher risk because its newer buildings are worth more.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A coastal nation ~~with a fast-growing tourism industry~~ sits where a dense oceanic plate sinks beneath a continental plate. ~~Its capital has a new subway system.~~ Which set of hazards should its planners expect?

Target word: hazards of this boundary **Decides:** Oceanic plate subducting under continental plate.

A	Adjacent category	That fits a transform boundary, not subduction.
B	Confusion set	That pattern fits divergent boundaries or hot spots.
C	Correct	Subduction zones produce all three.
D	Dropped element	Drops the earthquakes and volcanoes that come with subduction.

2. Answer B Concept 1 · transfer

A region has frequent shallow earthquakes along a long fault where two plates slide past each other horizontally, but it has no volcanoes. What kind of plate boundary is this?

Target word: boundary from its hazards **Decides:** Horizontal sliding, shallow quakes, no volcanoes.

A	Confusion set	Divergent boundaries pull apart and produce volcanism.
B	Correct	Plates slide sideways; no magma is produced.
C	Absolute language	Horizontal motion rarely lifts the seafloor enough for a tsunami.
D	Adjacent category	Subduction produces volcanoes and deep quakes.

3. Answer C Concept 2

A seismic network ~~with 120 stations~~ records a magnitude 7.0 earthquake. A mountain village near the epicenter reports heavy damage (intensity IX), while a city 300 km away ~~that hosts a university~~ reports light shaking (intensity IV). How should these numbers be understood?

Target word: magnitude vs. intensity **Decides:** One quake, two places.

A	Confusion set	Those are intensities, not magnitudes.
B	Dropped element	Drops distance and ground conditions, which change intensity.
C	Correct	Magnitude measures energy at the source; intensity measures effects at a site.
D	Absolute language	Intensity varies from place to place by definition.

4. Answer D Concept 2 · transfer

About how much more energy does a magnitude 8.0 earthquake release than a magnitude 6.0 earthquake?

Target word: energy ratio across two magnitude units **Decides:** About 32× per unit, compounded.

A	Inverted operation	Adds 32 + 32 instead of multiplying.
B	Wrong stem number	Uses the 2-unit difference as the ratio.
C	Confusion set	That is the ratio of ground-motion amplitude (10×10), not energy.
D	Correct	$32 \times 32 \approx 1,000$.

5. Answer B Concept 3

A volcano observatory ~~staffed by 15 scientists~~ monitors two volcanoes. Volcano X erupts low-silica basalt; Volcano Y erupts high-silica rhyolite rich in dissolved gas. ~~Volcano X is closer to the observatory's access road.~~ Which is more likely to erupt explosively?

Target word: which is explosive **Decides:** High-silica, gas-rich magma.

A	Absolute language	Composition and gas, not size, control explosivity.
B	Correct	High viscosity lets pressure build.
C	Adjacent category	Size (shield volcanoes) is not explosivity.
D	Inverted operation	Hotter basalt is runnier, so gas escapes easily and eruptions are gentle.

6. Answer D Concept 4

Residents of a fishing town ~~with a colorful harbor festival~~ feel a long, strong earthquake. Minutes later the sea pulls far back from the beach, exposing the seafloor. ~~A tour boat is scheduled to leave at noon.~~ What should residents do?

Target word: correct response **Decides:** Strong quake plus drawback.

A	Absolute language	A strong quake and drawback are reliable warning signs; waiting costs time.
B	Dropped element	Drops the fact that tsunamis come as a series; later waves can be larger.
C	Inverted operation	The water will return fast and powerfully.
D	Correct	The drawback is a natural warning that a tsunami is arriving within minutes.

7. Answer B Concept 5

A hurricane with 190 km/h winds, ~~named by the regional warning center~~, crosses 29 °C ocean water and then moves over a cold current of 22 °C. ~~Its forecast track passes near a major shipping port.~~ What will most likely happen to its strength?

Target word: change in strength **Decides:** Water below about 26 °C.

A	Inverted operation	The storm loses its heat source.
B	Correct	Hurricanes run on heat from warm sea surfaces.
C	Confusion set	Coriolis depends on latitude, not water temperature.
D	Absolute language	Intensity changes with the energy supply.

8. Answer A Concept 6

A town of 9,000 is built on the floodplain of a river whose 100-year flood reached the courthouse steps in 1998. ~~The town's bridge was repainted last year.~~ What is the probability that a flood at least that large occurs in any given year?

Target word: annual probability **Decides:** 100-year flood = 1 in 100 per year.

A	Correct	The return period is the inverse of the annual probability: 1/100.
B	Inverted operation	That is the chance of no such flood in a year.
C	Absolute language	It can happen twice in a decade or not at all in a century.
D	Confusion set	The return period is an average, not a schedule.

9. Answer D Concept 6 • transfer

A 50-year flood has a 2% chance of happening in any year. What is the chance of at least one such flood during a 30-year mortgage?

Target word: probability over a span of years **Decides:** $1 - 0.98^{30}$.

A	Wrong stem number	That is the chance for a single year.
B	Dropped element	Adds 2% thirty times, dropping the compounding of no-flood years.
C	Inverted operation	That is the chance of no flood.
D	Correct	$0.98^{30} \approx 0.55$ chance of none, so $\approx 45\%$ of at least one.

10. Answer C Concept 7

A county ~~that promotes its farmers' markets~~ has paved 40% of a once-forested watershed for new subdivisions over 20 years. The same size rainstorm now produces a higher and faster river crest. ~~The county seat's library was recently expanded.~~ Why?

Target word: why the flood peak rose **Decides:** Paving over forest.

A	Absolute language	Land cover strongly controls how rain becomes runoff.
B	Confusion set	Evaporation removes water; it does not raise crests.
C	Correct	More water arrives faster, raising the crest.
D	Inverted operation	Pavement blocks infiltration.

11. Answer A Concept 7 • transfer

A region that once had about one felt earthquake a year begins having dozens a year after large volumes of wastewater are injected deep underground. Which explanation fits best?

Target word: mechanism of induced seismicity **Decides:** Fluid injection near faults.

A	Correct	Higher pore pressure unclamps stressed faults.
B	Adjacent category	Plate boundaries form over millions of years, not from injection.
C	Absolute language	Induced earthquakes are well documented.
D	Inverted operation	The activity described adds fluid; it does not remove it.

12. Answer A Concept 8

Two towns sit the same distance from the same active fault and face the same expected shaking. Town P has 50,000 people in unreinforced brick buildings; Town Q has 50,000 people in buildings retrofitted to modern codes. ~~Town P has a famous clock tower.~~ Which statement is correct?

Target word: hazard vs. risk **Decides:** Same hazard and exposure, different vulnerability.

A	Correct	Risk = hazard $\times$ exposure $\times$ vulnerability.
B	Confusion set	The hazard (shaking) is the same; the difference is vulnerability.
C	Dropped element	Drops vulnerability from the risk equation.
D	Inverted operation	Retrofitting lowers vulnerability and therefore risk.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	4	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	8	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Plate boundary sets the hazard	Subduction zones produce megathrust earthquakes, tsunamis, and explosive volcanoes; divergent boundaries produce shallow quakes and gentle eruptions; transform boundaries produce shallow quakes without volcanoes.	Treating all plate boundaries as having the same hazards.	The type of plate motion described in the stem.
2	Magnitude vs. intensity	Magnitude is one number for the energy an earthquake releases; intensity is the shaking and damage at a place and varies with distance and ground. Each whole magnitude step is about 32 times more energy.	Thinking each magnitude step is 10 times the energy, or that one quake has one intensity.	One quake with different damage reports at different places.
3	Silica and gas control explosivity	High-silica, gas-rich magma is viscous and traps gas, so it erupts explosively; low-silica basalt is runny and erupts gently.	Thinking hotter or bigger volcanoes are the most explosive.	Magma composition named in the stem.
4	Tsunami behavior and warning signs	Tsunamis come from sudden vertical seafloor movement, arrive as a series of waves, and often begin with the sea drawing back; the right response is to move to high ground at once.	Treating the drawback as safe or the first wave as the only wave.	A strong quake near the coast followed by the sea retreating.
5	Hurricane fuel	Hurricanes draw energy from warm ocean water (about 26 °C or warmer) and weaken over land or cooler water; storm surge is often the deadliest hazard.	Thinking storm strength is fixed once a hurricane forms.	A storm moving onto land or over cool water.
6	Recurrence interval and probability	A 100-year flood has a 1% chance of happening in any year; the chance over n years is $1 - (1 - p)^n$.	Reading “100-year” as a schedule, or multiplying p by n.	A named return period and a question about a year or a span of years.
7	Human activity changes hazard frequency	Paving increases runoff and flood peaks; deep wastewater injection can trigger earthquakes by raising fluid pressure on faults.	Assuming natural hazards are unaffected by people.	A change in land use or subsurface activity followed by a change in events.
8	Hazard vs. risk	Risk combines the hazard with exposure (who and what is in harm’s way) and vulnerability (how easily they are harmed).	Equating the same hazard with the same risk.	Same event threat, different buildings or populations.

Course scope (public description): Natural Hazards delves into the complex and dynamic processes that govern Earth's natural systems and the hazards they produce. Students will explore how the geosphere, atmosphere, hydrosphere, and biosphere interact to create natural events such as earthquakes, volcanic eruptions, hurricanes, floods, and tsunamis. The course emphasizes the role of plate tectonics in generating lithospheric hazards, analyzing how movements within the Earth's crust lead to earthquakes and volcanic activities. Additionally, students will learn about the formation of atmospheric hazards, including how climate patterns and atmospheric circulation contribute to hurricanes, floods, and tornadoes. Students will examine how human activities impact the frequency and intensity of natural hazards and how natural hazards influence human behavior. By the end of the course, students will have a comprehensive understanding of the scientific principles underlying natural hazards and will be equipped to analyze and address the challenges posed by these events in both natural and human-altered environments.